MIL-C-81983(AS)
24 September 1974
SUPERSEDING
AS 1083
15 December 1969

MILITARY SPECIFICATION

CARTRIDGE, DELAY, MARK 142 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly, and preparation for delivery of the Mark 142 Mod 0 delay cartridge and the methods of inspection and tests upon which product acceptance will be based. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-20444	Primer, Percussion, M42, Loading, Assembling and Packing
MIL-M-21383	Manganese Delay Composition
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-22264	Powders, Ignition, Gasless

FSC 1377

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS AND ASSOCIATED LISTS

Naval Air Systems Command (Code Ident 30003)

DL 2519943	Cartridge, Delay, Mark 142 Mod 0, List of Drawings and Specifications
DL 2520036	Test Fixture, Cartridge, Delay, Mark 142 Mod 0
2519943	Cartridge, Delay, Mark 142 Mod 0 Assembly
2519945	Insert Assembly
2838308	Cartridge Packaging Assembly
2838309	Container, Inner, Assembly

Ordnance Corps of the Army (Code Ident 19203)

8799925	Primer, M42, Assembly and Details

PUBLICATIONS

Naval Ordnance Systems Command (Code Ident 10001)

OP 5	Ammunition Ashore (This Publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)
WR-43A	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

49 CFR 171-178 Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The Mark 142 Mod 0 delay cartridge covered by this specification shall be manufactured in accordance with DL 2519943 and all documents listed thereon.

3.2 **Materials.** All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 **Preproduction sample.** Unless otherwise specified by the procuring activity, a preproduction sample of 82 Mark 142 Mod 0 delay cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 **Primary components.** For the purposes of this specification the primer, M42, Drawing 8799925; the gasless ignition powder, MIL-P-22264; the delay composition, MIL-M-21383; and the main propellant charge, Drawing 2519943, are defined as primary components. (See 4.3.3.)

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3.4 **Primer.** The M42 percussion primer used in the Mark 142 Mod 0 delay cartridge shall be manufactured in accordance with Drawing 8799925, loaded with primer mixture number 793, and accepted in accordance with the requirements of MIL-P-20444. The primers shall be furnished from lots manufactured within the preceding 12 months. (See 4.7.1.)

3.4.1 **X-ray.** Primers shall be free of the following defects when examined by X-ray in accordance with 4.7.2: missing or abnormally thin charge; charge broken, cracked or porous; cracked or off-center anvil, or other misplaced or deformed inert parts; anvil abnormally close to primer cup dome; and all other observable defects in assembly.

3.5 **Propellant charge.**

3.5.1 **Quantity.** All propellant charges shall be as specified on Drawing 2519943. Charges shall be determined by weight rather than by volume measurements.

3.5.2 **Smokeless powder, handling and loading.** Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.5.3 **Smokeless powder condition.** The smokeless powder utilized shall exhibit no evidence of decomposition. (See 4.7.3.)

3.6 **Gasless ignition powder.** The A-1A gasless ignition powder shall be accepted in accordance with the provisions of MIL-P-22264 and loaded into the cartridge as specified on Drawing 2519945.

3.7 **Manganese delay composition.** The manganese delay composition shall be accepted in accordance with the requirements of MIL-M-21383 and loaded into the cartridge as specified on Drawing 2519945.

3.8 **Cartridge inspection.**

3.8.1 **Visual inspection.** Cartridges shall be free of the following visible defects: burrs, dents, deep scratches, split or cracked edges, chipping or flaking of the protective finish, sharp edges, defective sealant application around the crimped area; and all other defects

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which could conceivably prevent entry into the firing chamber or adversely affect ballistic performance. (See 4.7.4.)

3.8.2 **Leakage.** Cartridge leakage rate shall not exceed 10^{-5} cc of air or gas per second when tested in accordance with 4.7.5.

3.8.3 **X-ray.** Cartridges shall be free of imperfections in assembly when examined by X-ray in accordance with 4.7.6.

3.8.3.1 **Serial number identification.** Cartridges shall be legibly identified by serial numbers in accordance with 4.7.6.

3.9 **Ballistic requirements.** Cartridges from each production lot, selected in accordance with 4.7.7.1, shall meet the requirements listed below when subjected to the test prescribed in 4.7.7.1.

3.9.1 **Delay time.** The upper and lower limits of delay time shall be 0.175 second and 0.075 second, respectively.

3.9.2 **Velocity.** The upper and lower limits of average slug velocity shall be 330 ft/sec and 230 ft/sec, respectively.

3.9.3 **Misfire.** There shall be no misfires. (See 4.7.7.2.1.)

3.10 **Workmanship.** Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes, coatings, and quality of crimping operations.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Sampling.** Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105.

4.3 **Lots.**

4.3.1 **Inspection lots.** Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2 **Preproduction sample.** Before entering into quantity production, an acceptable preproduction sample of 82 Mark 142 Mod 0 delay cartridges shall be prepared. Of these cartridges, 72 shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of 3.3 through 3.8 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625: 40-foot drop test (6); 28-day temperature and humidity cycling (THC) (18); and vibration (12). The number in parentheses denotes the number of cartridges allocated for test. These cartridges (with exception of the 40-foot drop cartridges), after exposure to environmental treatments, shall be conditioned for a minimum of 6 hours at 70° ± 5° F together with 12 untreated cartridges and fired in the test fixture of 4.7.7.1. In addition, 24 cartridges shall be temperature conditioned for a minimum of 6 hours (12 at -65° ± 5° F and 12 at 160° ± 5° F) and fired in the test fixture of 4.7.7.1. All cartridges shall fire and produce results within the limits set forth in 3.9.1 and 3.9.2 except that the upper limit of delay time shall be 0.250 second for the sample subjected to THC. The failure of any cartridge to meet the acceptance requirements as stated above shall cause rejection of the preproduction sample. Any production prior to acceptance of the preproduction sample shall be at the supplier's risk.

4.3.3 **Production lot.** Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 501 and 1200 cartridges, respectively. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4 **Gages.** The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

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4.5 **Classification of characteristics.** The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43A. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 **Quality conformance inspection.** Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 2519943 and verification of the following requirements and tests of table I.

Table I

TEST AND REQUIREMENT INDEX

Item	Nature of requirement verification	Applicable paragraphs	Classification
Primary component	Single lot	3.3, 4.3.3	(M101)
Primer	Age, loading and acceptance	3.4, 4.7.1	(C1)
Primer	X-ray	3.4.1, 4.7.2	(C2)
Propellant	Type and weight	3.5.1	(C3)
Smokeless powder	Condition	3.5.3, 4.7.3	(C4)
Ignition powder	Loading and acceptance	3.6	(C5)
Delay composition	Loading and acceptance	3.7	(C6)
Cartridge	Visual inspection	3.8.1, 4.7.4	(M102)
Cartridge	Leakage	3.8.2, 4.7.5	(C7)
Cartridge	X-ray	3.8.3, 4.7.6	(C8)
Cartridge	Serial number identification	3.8.3.1, 4.7.6	(M103)
Cartridge	Ballistic requirements	3.9, 4.7.7	(C9)
Cartridge	Packaging, packing and container markings	4.6.1	(M104)

4.6.1 **Packaging, packing and marking.** The packaging and packing of cartridges and the container markings shall be in accordance with section 5. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in section 5.

4.7 Tests.

4.7.1 **Primer verification.** Verification shall be provided that the primers used were manufactured, loaded, and accepted in accordance with the requirements of 3.4.

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4.7.2 **Primer X-ray.** Examine all primers by X-ray and reject those having any of the defects of 3.4.1. Pressure-sensitive tape may be used as an aid in positioning the primers in suitable grooved racks. For the most revealing exposure, the primers are laid on their side with the anvil cutouts vertical.

4.7.3 **Smokeless powder condition.** Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.7.4 **Visual inspection.** Examine all cartridges and reject those having any of the defects of 3.8.1.

4.7.5 **Leakage.** Each cartridge in the preproduction sample or production sample shall be leak tested in a dry gas or dry air leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cc/sec gas or air at a pressure differential of 1 atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.6 **X-ray examination.** Examine all cartridges by X-ray and any observable imperfections in assembly shall be cause for cartridge rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to X-ray examination. The cartridge serial numbers shall be in consecutive order beginning with the number one (1) in each production lot. The cartridges for X-ray shall be arranged on trays or boards in consecutive numerical order and each X-ray plate shall carry a permanent identification of the cartridges displayed thereon. The X-ray plate identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. The serial numbers shall be rubber stamped on the cartridges with approximately one-eighth inch high characters using the same process as that applying to the other cartridge markings. The serial numbers shall be located apart from the other cartridge markings. X-ray plates of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. (See 6.2.)

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4.7.7 **Functional test.**

4.7.7.1 The number of cartridges selected from each production lot for the functional test shall be in accordance with the requirements of MIL-STD-105, Inspection Level II, for an AQL of 0.15, single sampling plan for normal inspection. The cartridges shall be conditioned at 70° ± 5° F for a minimum of 6 hours and fired in the test fixture, DL 2520036. Instrumentation shall be provided to measure the delay time which is defined as the time interval from sear removal from the firing mechanism to first indication of slug movement, and to measure the average slug velocity over an interval of 24.00 ± 0.06 inches commencing 36.0 ± 0.5 inches from the test fixture muzzle. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from the production lot for investigative purposes.

4.7.7.2 **Criteria of acceptability.** Cartridges of the functional sample shall meet the requirements of 3.9. Acceptance for the delay time requirement of 3.9.1 and the velocity requirement of 3.9.2 shall be in accordance with MIL-STD-105 using an AQL of 0.15 applied separately to the requirement of each paragraph.

4.7.7.2.1 **Misfire.** The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.8 **Retest.** There shall be no retests.

4.8.1 **Test failure.** If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and that portion of the test reconducted.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.** For definitions of packaging levels see 6.4.

5.1.1 **Level A.** Level A packaging shall be used for packaging of all production lots for service use. Two cartridges shall be packaged in an inner container in accordance with Drawing 2838309.

5.1.2 **Level C.** When directed by the procuring activity, cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

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5.2 **Packing.** For definitions of packing levels see 6.4.

5.2.1 **Level A.** Level A packing shall be used for packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with Drawing 2838308.

5.2.2 **Level C.** When directed by the procuring activity, cartridges intended for expenditure in preproduction sample testing and for production lot acceptance tests, and are packaged in accordance with 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 **Markings.**

5.3.1 **Special markings.** Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 **Normal markings.** Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on Drawings 2838308 and 2838309. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 **Intended use.** The Mark 142 Mod 0 delay cartridge was designed to actuate the personnel parachute thruster, Propellex P/N 163-3, during low speed, low altitude emergency ejections from the OV-10A aircraft.

6.2 **Ordering information.** Procurement documents should specify the following:

a. Title, number and date of this specification

b. Whether preproduction inspection is required, and if so, specify the test activity

c. Production lot size and test activity (see 4.3.3)

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d. Direction for shipping X-ray plates of entire production lot along with ballistic sample (4.3.3) to the activity conducting production lot acceptance test

e. Items of data required for each preproduction and production lot (see 6.3)

f. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore," OP 5, are applicable.

6.3 **Data.** For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 **Test report.** Results of ballistic tests shall be forwarded to the Naval Ordnance Station (5012), Indian Head, Md., and the procuring activity.

6.3.2 **Data cards.** The contract or order should state the format, quantity, and destination of data cards.

6.4 **Definitions.** For the purpose of this specification, the following definitions apply.

6.4.1 **Level A packaging.** The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during handling, shipment, indeterminate storage, and world-wide redistribution.

6.4.1.1 **Level C packaging.** The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 **Level A packing.** The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage, and world-wide redistribution.

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6.4.2.1 **Level C packing.** The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirement of this level.

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-81983(AS) CARTRIDGE, DELAY MK 142 Mod 0

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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